

Offline and Preference-Based Reinforcement Learning for Robotics: A Comprehensive Survey

Abstract

The deployment of reinforcement learning (RL) in real-world robotics has historically been hindered by the prohibitive costs of online exploration, safety constraints, and the difficulty of specifying dense reward functions. Recently, the convergence of Offline Reinforcement Learning (Offline RL) and Preference-Based Reinforcement Learning (PbRL) has emerged as a transformative paradigm, enabling agents to learn complex behaviors from static datasets augmented with sparse, relative human feedback. This survey provides a comprehensive review of the state-of-the-art in offline and preference-based RL for robotics. We synthesize over 300 recent studies to construct a detailed taxonomy of methodological families, ranging from reward modeling and direct policy optimization to implicit preference learning and multi-modal feedback integration. We examine techniques for learning from suboptimal demonstrations, handling noisy human labels, and leveraging foundation models for scalable reward inference. Furthermore, we analyze critical challenges including distributional shift, credit assignment in long-horizon tasks, safety alignment, and the evaluation bottlenecks inherent to human-in-the-loop systems. By integrating theoretical insights with empirical evidence from diverse robotic domains—including manipulation, locomotion, and human-robot interaction—this paper outlines the current landscape and identifies pivotal directions for future research.

1. Introduction

Reinforcement Learning (RL) offers a principled framework for training autonomous agents to maximize cumulative rewards through interaction with an environment. However, applying standard online RL to physical robotics presents significant barriers: the sample inefficiency of model-free algorithms often requires millions of interactions [1], which is infeasible due to hardware wear and safety risks [2]; and the engineering of dense, task-specific reward functions is notoriously difficult, often leading to reward hacking or unintended behaviors [3].

To address these challenges, the field has increasingly turned to **Offline Reinforcement Learning**, which learns policies from fixed datasets without further environment interaction [4], and **Preference-Based Reinforcement Learning (PbRL)**, which replaces explicit reward engineering with relative human feedback on trajectory segments [5]. The intersection of these two fields—learning from offline data guided by human preferences—has catalyzed a new wave of research capable of extracting robust policies from suboptimal, heterogeneous datasets while aligning robot behavior with complex human values [6].

The evolution of this field can be characterized by a shift from purely demonstrative learning (Imitation Learning) to evaluative learning. Early approaches relied heavily on Behavioral Cloning (BC), which suffers from compounding errors and an inability to surpass the quality of the demonstration data [4]. Offline RL methods like Conservative Q-Learning (CQL) and Implicit Q-Learning (IQL) mitigated distributional shift by penalizing out-of-distribution actions [7], yet they still struggled with the “reward problem”: how to define optimality in the absence of ground-truth rewards. PbRL addresses this by learning a reward function from pairwise comparisons [8], but traditional PbRL requires extensive online exploration to generate queryable trajectories.

Recent advancements have unified these paradigms. Methods now leverage offline datasets to pre-train dynamics models or policies, using human preferences solely to refine reward functions

or select among candidate behaviors [9]. Innovations include learning rewards from suboptimal rankings [10], utilizing vision-language models (VLMs) to automate preference labeling [11], and developing direct policy optimization algorithms that bypass explicit reward modeling entirely [12]. Furthermore, the integration of implicit feedback signals, such as EEG-based error potentials [13] and physical corrections [14], has expanded the modalities through which humans can guide robotic learning.

This survey synthesizes the extensive literature on offline and preference-based RL for robotics. We organize the field into coherent methodological systems, analyzing how different approaches handle data heterogeneity, feedback efficiency, and safety. We cover the spectrum from theoretical foundations of identifiability [15] to practical deployments in surgical robotics [16] and assistive devices [17]. By critically evaluating over 300 distinct contributions, we aim to provide a definitive resource for understanding the mechanisms, limitations, and future potential of learning robotic behaviors from offline experience and human preference.

2. Method

The methodological landscape of offline and preference-based RL for robotics is diverse, encompassing techniques for reward inference, direct policy optimization, representation learning, and feedback integration. We categorize these approaches into several key families based on their core mechanisms for utilizing offline data and preference signals.

2.1 Reward Modeling from Offline Preferences

A dominant paradigm involves learning a reward function $\hat{r}_\theta$ from offline preference data, which is then used to train a policy via standard offline RL algorithms. This decoupling allows for the reuse of established offline RL solvers while leveraging the expressivity of human preferences.

2.1.1 Bradley-Terry Models and Active Querying The foundational approach models human preferences using the Bradley-Terry model, where the probability of preferring trajectory σ_1 over σ_2 is determined by the exponential of their cumulative rewards [8]. In offline settings, methods like **OPAL** (Offline Preference-based Apprenticeship Learning) select informative trajectory pairs from a static dataset using active learning strategies based on reward model uncertainty [6]. Similarly, **OPRL** formalizes fully offline preference learning by employing ensemble networks to estimate reward uncertainty and selecting queries that maximize information gain [18]. These methods demonstrate that offline datasets containing random or multi-task data provide sufficient signal for reward inference when paired with active learning, often matching ground-truth reward performance with as few as 15 queries [6][18].

To improve sample efficiency, **SENIOR** integrates motion-distinction-based selection using Kernel Density Estimation (KDE) and Principal Component Analysis (PCA) to identify low-density, high-distinction segment pairs, significantly improving label quality [19]. **DAPPER** further advances query efficiency by replacing single-policy comparisons with policy-to-policy queries, training a discriminator to estimate preference discriminability and mitigating policy bias [20].

2.1.2 Handling Noisy and Suboptimal Data Real-world offline datasets often contain noisy or suboptimal trajectories. **TROFI** (Trajectory-Ranked Offline Inverse RL) addresses this by training a reward model using T-REX on a subset of human-ranked trajectories, enabling effective

policy optimization on the remaining unlabeled, suboptimal data [21]. **RIME** (Robust Preference-based RL) introduces a denoising discriminator that utilizes dynamic KL divergence bounds to filter corrupted preferences and flip unreliable labels, demonstrating robustness even at error rates of 20% [22].

For datasets with mixed quality, **TREND** employs a cyclic tri-teaching strategy where three reward networks collaboratively identify clean samples via small-loss principles, integrating minimal expert demonstrations to anchor the learning process [23]. **MTPL** (Multi-Type Preference Learning) formulates reward learning as a multi-task problem, simultaneously optimizing for explicit ranking and “equal preference” (minimizing MSE for similar behaviors), which forces the reward model to assign similar values to indistinguishable segments, significantly improving performance in low-preference regimes [24].

2.1.3 Listwise and Scale Feedback Moving beyond binary comparisons, **LiRE** (Listwise Reward Estimation) constructs a Ranked List of Trajectories (RLT) via sequential binary search using ternary feedback, exploiting transitivity to infer second-order preferences (relative strength) without complex feedback types [25]. **R4** (Ranked Return Regression) treats ratings as ordinal targets via a differentiable sorting operator, minimizing a ranking mean squared error that preserves relative quality information while allowing flexible return distributions [26]. **Scale Feedback Learning** extends binary choices to continuous scale feedback via a slider, constraining the feasible weight space more tightly than discrete choices and reducing worst-case error bounds [27].

2.2 Direct Policy Optimization without Explicit Rewards

A growing body of work argues that explicit reward modeling introduces unnecessary error propagation. These methods optimize policies directly from preference data using contrastive objectives or implicit mappings.

2.2.1 Contrastive Preference Learning **CPL** (Contrastive Preference Learning) exploits the bijection between optimal advantage functions and policies in Maximum Entropy RL. It substitutes log-probabilities for advantages in the regret-based preference model, yielding a convex, supervised objective that bypasses reward modeling entirely [12]. **DPPO** (Direct Preference-based Policy Optimization) defines a policy-segment distance metric and constructs a contrastive score function that maximizes proximity to preferred segments while distancing from unpreferred ones, outperforming reward-modeling baselines on dexterous manipulation tasks [28].

PLARE leverages Vision-Language Models (VLMs) to generate preference labels for unlabeled trajectory segments, optimizing policies via supervised contrastive learning without an intermediate reward model. This approach mitigates error propagation and achieves higher success rates than VLM-based reward generation methods [29]. **IPL** (Inverse Preference Learning) eliminates explicit reward modeling by leveraging the inverse soft-Bellman operator to optimize a single Q-network that simultaneously satisfies the Bellman equation and preference constraints, achieving competitive performance with 10x fewer parameters [30].

2.2.2 Direct Preference Optimization (DPO) Variants Inspired by advances in language model alignment, **FDPP** fine-tunes diffusion policies using Proximal Policy Optimization (PPO) guided by a preference-based reward function, incorporating KL regularization to prevent reward overfitting [31]. **Diffusion-RKO** unifies Kahneman-Tversky Optimization (KTO) and Relative Preference Optimization (RPO), utilizing per-sample binary labels and context-aware reweighting

to focus learning on semantically ambiguous regions in deformable object manipulation [32]. **Flow-PRO** introduces RPRO, a preference objective combining contrastive optimization with an explicit proximal regularizer to anchor implicit reward magnitudes, preventing policy drift in flow-matching Vision-Language-Action (VLA) models [33].

2.3 Implicit and Latent Preference Learning

These methods encode preferences implicitly within latent representations or value functions, avoiding the need for explicit reward signals or pairwise comparisons during inference.

2.3.1 Hindsight and Future-Conditioned Rewards **HPL** (Hindsight Preference Learning) models rewards conditioned on future trajectory segments using a Variational Auto-Encoder (VAE) to compress hindsight information. By marginalizing over VAE-derived priors, it corrects bias in high-risk actions and distinguishes successful from failed trajectories [34]. **OPPO** (Offline Preference-guided Policy Optimization) jointly optimizes a contextual policy and a hindsight information extractor via an Offline Hindsight Information Matching (HIM) objective, preserving more task information in a high-dimensional latent context space than scalar rewards [35].

2.3.2 Representation-Aligned Learning **RAPL** (Representation-Aligned Preference-based Learning) decouples representation learning from reward modeling. It fine-tunes a visual encoder using human preference triplets to align robot embeddings with human latent spaces, subsequently constructing dense visual rewards via Optimal Transport. This allows simple feature-matching rewards to distinguish preferred behaviors effectively [36][37]. **SIRL** (Similarity-based Implicit Representation Learning) utilizes human-labeled trajectory triplets to minimize a symmetric triplet loss, forcing embeddings of perceptually similar trajectories to cluster, thereby encoding causal features independent of specific reward weights [38].

2.3.3 Behavior Preference Regression **BPR** (Behavior Preference Regression) reformulates policy improvement as a least-squares regression problem on paired action samples, fitting the maximum mode of the Q-function while maximizing behavioral consistency by regressing the log-ratio of policy densities [39]. **PAO-DP** (Preferred-Action-Optimized Diffusion Policy) replaces weighted regression with preference optimization in diffusion policies, automatically generating preferred actions via critic-guided importance sampling [40].

2.4 Leveraging Foundation Models and Large Language Models

The integration of Large Language Models (LLMs) and Vision-Language Models (VLMs) has enabled scalable, automated preference labeling and reward design.

2.4.1 VLM-Based Preference Labeling **RL-VLM-F** queries VLMs using a two-stage prompting strategy (visual analysis followed by discrete label extraction) to generate preference labels for image pairs, significantly reducing noise compared to raw score querying [11]. **PrefVLM** utilizes a frozen VLM with learnable projection layers to generate coarse trajectory-level preferences, refined by sparse human feedback and a self-supervised inverse dynamics loss [41]. **PRISM** integrates a 3D Point Cloud-Language Model with a self-ranked Chain-of-Thought module to generate viewpoint-invariant geometric embeddings for preference-based RL [42]. **VARP** augments final-state images with 2D trajectory sketches to provide temporal context for VLM preference queries, improving accuracy by 16% over final-image-only baselines [43].

2.4.2 LLM-Assisted Reward Design and Feedback **LAPP** integrates LLMs into the RL feedback loop to generate preference labels for raw state-action trajectories based on high-level language instructions, enabling dense preference signals without visual inputs [44]. **CriticGPT** fine-tunes LLaVA via LoRA to analyze robotic trajectories and generate preference labels using Chain-of-Thought reasoning, automating the reward labeling process [45]. **Lafite-RL** uses an LLM as a real-time evaluator within the RL loop, interpreting scene descriptions to provide dense, phase-aware evaluative feedback [46]. **PROF** generates executable Python reward codes via LLM prompting and optimizes them using Reward Preference Ranking (RPR) based on expert vs. noisy trajectory returns [47].

2.4.3 Crowdsourced and Multi-Agent LLMs **PrefCLM** employs a collective of LLM agents to evaluate trajectories, aggregating scores using Dempster-Shafer Theory to resolve conflicts and mitigate individual reasoning errors [48]. **ROVED** utilizes a lightweight Vision-Language Embedding (VLE) to generate initial segment-level preferences, filtering samples via KL divergence to target oracle queries only on uncertain sets, reducing human query requirements by 80% [49].

2.5 Integrating Heterogeneous and Implicit Feedback

Human feedback is not limited to explicit preferences; it includes physical corrections, neural signals, and multimodal interactions.

2.5.1 Physical Corrections and Interventions **OHP-RL** integrates human interventions as online pairwise preference signals within an asynchronous actor-critic framework, using a state-dependent preference gate to modulate influence [50]. **PACT** constructs zero-cost preference pairs from human corrective actions versus policy actions at intervention points, leveraging a self-supervised progress model to localize suboptimal segments [51]. **MILE** constructs a differentiable probit-based intervention model estimating human takeover probability based on the value gap, utilizing non-intervention timesteps as implicit confirmation of optimality [52]. **HAPO** employs Prospect Theory to formulate a utility function from binary desirability signals derived from human interventions, enabling preference alignment without explicit pairing [53].

2.5.2 Neural and Physiological Signals **RLIHF** integrates pre-trained EEGNet decoders to transform raw EEG epochs into probabilistic error-related potential (ErrP) estimates, providing continuous, semantically meaningful evaluative feedback without interrupting task execution [13][54]. **EMPATHIC** maps temporal sequences of facial action units and head-motion features to reward probability distributions, formalizing Learning from Implicit Human Feedback (LIHF) [55]. **Neuroadaptation** frameworks utilize EEG-derived cognitive conflict signals to adjust robot admittance parameters in real-time, closing the loop between human cognitive state and robot control [56].

2.5.3 Multimodal and Combined Feedback **Combined Meta-Modalities Framework** integrates six distinct feedback types (Guidance, Correction, Demonstration, Exploration, Speed, Fallback) atop a preference-based algorithm, significantly reducing trials to first success [57]. **Unified Physical Interaction Learning** trains an ensemble of neural networks using a unified loss function comparing human inputs (demonstrations, corrections, preferences) against noisy trajectory deformations [14]. **DemPref** initializes a belief distribution over reward weights using passive demonstrations before refining via active preference queries, proving that passive data accelerates convergence [58][59].

2.6 Offline-to-Online Fine-Tuning and Hybrid Approaches

Many practical applications require initializing from offline data and fine-tuning with limited online interaction and preferences.

2.6.1 Confidence Sets and Constrained Exploration **BRIDGE** constructs an offline confidence set via a Hellinger ball around a BC policy estimator, constraining online preference learning exploration to this set to ensure bounded regret [60]. **PRC** (Preference-based RL with Constrained actions) executes a two-step framework where the second step constructs a clipped action space containing only actions where the behavior policy probability exceeds a threshold, preventing evaluation on out-of-distribution states [61]. **UGES** (Uncertainty-Guided Expert Sampling) dynamically switches between sub-optimal and expert buffers based on Q-ensemble variance thresholds, strategically injecting human demonstrations where uncertainty is high [62].

2.6.2 Model-Based Hybrid Learning **MOTO** employs a recurrent VAE world model trained on offline data, utilizing model-based value expansion to estimate returns from latent rollouts, mitigating representation shift during online fine-tuning [63]. **H2O** (Hybrid Offline-and-Online RL) integrates real-world offline data with imperfect simulator rollouts via dynamics-aware policy evaluation, using discriminators to estimate state-action dynamics gaps and adaptively weight value regularization [64]. **Sim-OPRL** learns transition and reward ensembles from offline data, selecting pairs of exploratory policies maximizing preference uncertainty within a pessimistic transition model [9].

2.6.3 Skill Discovery and Hierarchical Approaches **S-EPOA** executes unsupervised skill pretraining to generate diverse behavioral primitives, using a trajectory estimator to predict skill returns and selecting queries that maximize behavioral distinguishability [65]. **PIPER** employs a two-level hierarchy where a high-level policy predicts subgoals, replacing human feedback with Primitive-in-the-Loop (PiL) signals derived from environment sparse rewards to train a high-level reward model [66]. **HPO** (Hierarchical Preference Optimization) formulates HRL as a bi-level optimization, deriving a primitive-regularized token-level DPO objective that incorporates lower-level value estimates to ensure subgoal feasibility [67].

2.7 Safety, Robustness, and Alignment

Ensuring that learned policies are safe and robust to distributional shifts is critical for robotic deployment.

2.7.1 Safe Offline RL with Preferences **PreSa** (Preference and Safety Alignment) integrates preference alignment and safety alignment into a unified constrained optimization problem, utilizing binary safety labels to define a feasible policy set via a sigmoid-based loss derived from Prospect Theory [68]. **SAFER** infers a latent safety variable from state windows using a Transformer-based posterior, employing a contrastive loss to distinguish safe versus unsafe demonstrations in latent space [69]. **PAINT** adapts reward sketching for continuous glucose monitoring, employing safety-constrained offline RL tuning using a verifiable prior to ensure glycemic safety [70].

2.7.2 Robustness to Corruption and Noise **Corruption Robust Offline RLHF** partitions corrupted datasets for robust reward estimation, solving trimmed maximum likelihood estimation

under Huber contamination to learn reward parameters even with adversarial noise [71]. **Best-of-Venom** highlights vulnerabilities by injecting poisoned preference pairs, demonstrating that even 1-5% poisoned data can bias Reward Models, necessitating robust training strategies [72]. **RIME** and **TREND** (discussed in 2.1.2) specifically address noisy labels through denoising discriminators and tri-teaching mechanisms [23][22].

2.7.3 Value Alignment and Personalization **PEARL** employs Gromov-Wasserstein distance to align trajectory distributions between source and target tasks, enabling zero-shot transfer of preference labels for personalized manipulation [73]. **Promptable Behaviors** trains a single policy conditioned on randomly sampled reward weight vectors, inferring low-dimensional weight vectors from group trajectory comparisons to personalize multi-objective rewards [74]. **CHARM** employs a Multilayer Perceptron to predict scalar feedback values and delays based on human characteristic questionnaire responses, integrating personality factors into feedback modeling [75].

3. Challenges and Outlook

Despite significant progress, several critical challenges remain in deploying offline and preference-based RL for robotics.

3.1 Data Quality and Distributional Shift

The performance of offline RL is heavily dependent on the coverage and quality of the dataset. Methods often fail when the offline data lacks sufficient diversity or contains high proportions of suboptimal trajectories [1][76]. While techniques like **CQL** [7] and **IQL** mitigate extrapolation error, the “compositionality gap” remains: standard algorithms struggle to stitch together sub-segments of optimal behavior from suboptimal trajectories to solve novel tasks [76]. Furthermore, cross-embodiment learning—transferring policies across robots with different morphologies—introduces gradient conflicts that static grouping strategies may fail to resolve [77]. Future work must focus on developing algorithms that can effectively extract compositional structures from heterogeneous, multi-robot datasets.

3.2 Feedback Efficiency and Human Burden

Although PbRL reduces the need for reward engineering, it shifts the burden to human labelers. Active querying strategies like **SENIOR** [19] and **DAPPER** [20] improve efficiency, but the cognitive load of distinguishing subtle trajectory differences remains high. Interfaces play a crucial role; **VR interfaces** have been shown to yield higher user confidence but induce distinct preference distributions compared to 2D views, complicating data aggregation [78]. Moreover, human feedback is often inconsistent, noisy, or influenced by cognitive biases [79]. Methods like **STEADY** attempt to stabilize scalar feedback dynamics [80], and **ABLUF** models trainer uncertainty [81], but robustly handling irrational or drifting human preferences remains an open problem.

3.3 Credit Assignment in Long-Horizon Tasks

Assigning credit to specific actions based on trajectory-level preferences is inherently difficult, especially for long-horizon tasks. **Preference Transformers** [82] and **PrefMMT** [83] use attention mechanisms to model non-Markovian rewards, yet they struggle when feedback is extremely sparse. **Search-Based Preference Weighting (SPW)** attempts to derive stepwise importance weights

by matching preference transitions to expert demonstrations [84], and **StARe** decomposes trajectories into semantic stages for stage-wise optimization [85]. However, automatic discovery of temporal structures without manual segmentation remains a significant hurdle.

3.4 Safety and Verification

Safety in offline PbRL is challenging because the agent cannot explore to verify safety constraints. **PreSa** [68] and **SAFER** [69] offer promising directions by integrating safety constraints directly into the optimization objective. However, ensuring that policies generalized from offline data do not exhibit unsafe behaviors in novel situations requires rigorous off-policy evaluation (OPE). **OPEHF** reconstructs immediate human rewards from sparse episode-level feedback to improve OPE estimators [86], but reliable safety guarantees without online interaction are still elusive. The risk of reward hacking, where agents exploit flaws in the learned reward model, persists, particularly when using powerful foundation models that may hallucinate preferences [33].

3.5 Evaluation Bottlenecks

The field lacks standardized benchmarks that accurately reflect real-world complexity. Many evaluations rely on simulated environments (e.g., MuJoCo, Meta-World) with synthetic preference labelers, which may not capture the noise and ambiguity of real human feedback [87][23]. Real-world benchmarks like **TOTO** [88] and **D5RL** [1] are emerging, but they often lack the scale and diversity needed to thoroughly test generalization. Furthermore, evaluating “alignment” is subjective; metrics like success rate do not capture nuanced human preferences regarding style, comfort, or social norms [89]. Future research must prioritize the development of comprehensive, real-world benchmarks with diverse human feedback modalities.

3.6 Scalability and Computational Cost

Integrating large foundation models (LLMs/VLMs) for reward labeling introduces significant computational overhead and latency [44][46]. While methods like **ROVED** [49] and **PrefCLM** [48] aim to reduce query costs, the inference time of large models can be prohibitive for real-time robotic control. Additionally, training complex architectures like **PrefMMT** [83] or **PRISM** [42] requires substantial GPU resources. Efficient distillation of these large models into lightweight robotic controllers is a critical direction for practical deployment.

4. Conclusion

The synergy between Offline Reinforcement Learning and Preference-Based Reinforcement Learning represents a pivotal advancement in robotic learning, offering a pathway to train capable, aligned agents without the risks of extensive online exploration or the brittleness of hand-crafted rewards. This survey has synthesized a vast array of methodologies, from robust reward modeling and direct policy optimization to the integration of implicit neural feedback and foundation models.

Key trends indicate a move towards **hybrid systems** that leverage offline data for initialization and online preferences for refinement, **direct optimization** techniques that bypass explicit reward modeling to reduce error propagation, and the **automation of feedback** via VLMs and LLMs. However, significant challenges persist in handling distributional shift, ensuring safety without online verification, and reducing the cognitive burden on human teachers.

Future research must address the scalability of these methods to complex, unstructured environments and develop rigorous theoretical guarantees for safety and convergence. The integration of multimodal human feedback, including physical corrections and physiological signals, holds promise for creating more intuitive and responsive human-robot teams. As the field matures, the focus must shift from algorithmic novelty on simulated benchmarks to robust, deployable systems that can learn safely and efficiently from the rich, heterogeneous data available in the real world. The convergence of these technologies suggests a future where robots can continuously adapt to human preferences, learning complex skills from static datasets and subtle human cues alike.

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